WEEK 10: LINEAR PROGRAMMING II — DUALITY

Advanced Algorithms

ROADMAP

1. Certifying an optimum: bounding an LP by combining its constraints
2. The dual LP, and a recipe for writing it
3. Weak duality (proved) and strong duality (stated)
4. Max-flow = min-cut, round two: as LP duality
5. The matching capstone: Kőnig's theorem
6. Complementary slackness — and what a negative price tells you

PART I — CERTIFICATES AND BOUNDS

HOW DO WE *CERTIFY* AN LP OPTIMUM?

Last week we saw an optimum of a (feasible, bounded) LP occurs at a vertex. A skeptic asks: *how do I know no feasible point does better?*

We have played this game before:

- **Week 8:** to certify a flow is maximum, exhibit a *cut* of the same value — no need to re-run Ford–Fulkerson.

HOW DO WE *CERTIFY* AN LP OPTIMUM?

Last week we saw an optimum of a (feasible, bounded) LP occurs at a vertex. A skeptic asks: *how do I know no feasible point does better?*

We have played this game before:

- **Week 4:** greedy picks k disjoint activities. For half-open intervals $[s, f)$, choose one point sufficiently close before each selected finish time. These k points *hit every activity* — so no $k + 1$ disjoint activities exist.

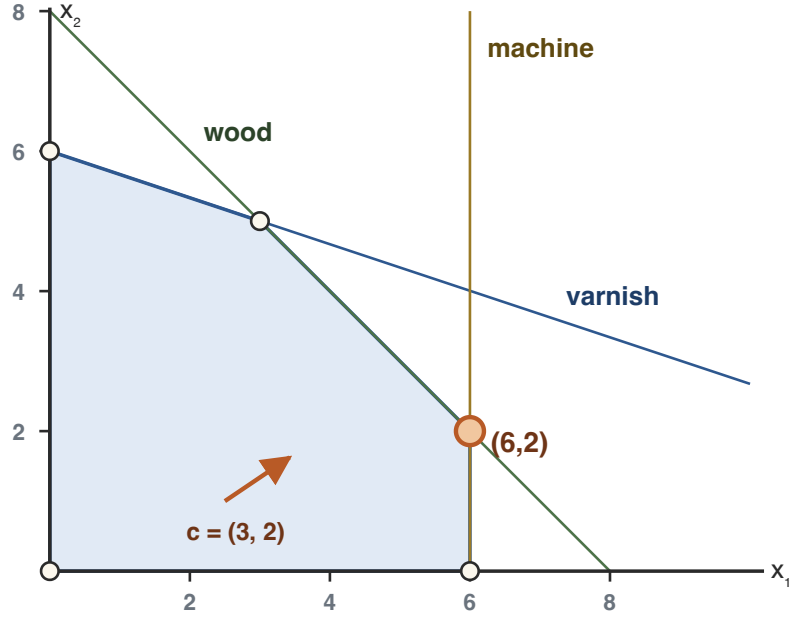
HOW DO WE *CERTIFY* AN LP OPTIMUM?

GOAL A short, checkable *certificate of optimality* for any LP.
Today's answer: a solution to a second LP — the **dual**.

LAST WEEK'S LP

$$
\begin{aligned}
\max \quad & 3x_1 + 2x_2 \\
\text{s.t.} \quad & x_1 \leq 6 \quad \text{(machine)} \\
& x_1 + x_2 \leq 8 \quad \text{(wood)} \\
& x_1 + 3x_2 \leq 18 \quad \text{(varnish)} \\
& x_1, x_2 \geq 0
\end{aligned}
$$

Chairs x_1, tables x_2.



The feasible region and the claimed optimum $(6, 2)$.

LAST WEEK'S LP

Same LP as Week 9, with (x, y) renamed (x_1, x_2) — the letter y is about to get a new job.

The sweep found $(6, 2)$, value $3 \cdot 6 + 2 \cdot 2 = 22$.

But *checking all vertices* is not a certificate that scales. We want an upper bound proof.

BOUNDING FROM ABOVE

$$\max\ 3x_1 + 2x_2 \quad \text{s.t.}$$

$$\underbrace{x_1 \le 6}_{\text{machine}},\quad \underbrace{x_1 + x_2 \le 8}_{\text{wood}},\quad \underbrace{x_1 + 3x_2 \le 18}_{\text{varnish}},\quad x_1, x_2 \ge 0$$

Idea: **combine the constraints**. Multiply each by a number $y_i \ge 0$ and add. If the result dominates the objective coefficient-wise, its right-hand side is an upper bound.

Attempt 1 — take $2\times$ (machine) $+\ 1\times$ (varnish):

$$2x_1 + (x_1 + 3x_2) \;=\; 3x_1 + 3x_2 \;\le\; 12 + 18 \;=\; 30.$$

Since $x_2 \ge 0$: $3x_1 + 2x_2 \le 3x_1 + 3x_2 \le 30$. Bound: 30.

BOUNDING FROM ABOVE

$$\max\ 3x_1 + 2x_2 \quad \text{s.t.}$$

$$\underbrace{x_1 \le 6}_{\text{machine}},\quad \underbrace{x_1 + x_2 \le 8}_{\text{wood}},\quad \underbrace{x_1 + 3x_2 \le 18}_{\text{varnish}},\quad x_1, x_2 \ge 0$$

Attempt 2 — take $3\times$ (wood):

$$3x_1 + 3x_2 \le 24.$$

Same left side as Attempt 1, cheaper right side. Bound: 24.

Getting closer to 22. Both attempts overshoot the x_2 coefficient ($3 > 2$). Can a mix avoid that waste?

YOUR TURN: A BOUND OF 22

$$
\begin{aligned}
x_1 &\le 6 && \text{(machine)} \\
x_1 + x_2 &\le 8 && \text{(wood)} \\
x_1 + 3x_2 &\le 18 && \text{(varnish)}
\end{aligned}
$$

CHALLENGE Combine these constraints with nonnegative weights to prove $3x_1 + 2x_2 \le 22$ for every feasible point.

Hint: try using just machine and wood. What weights give exactly $3x_1 + 2x_2$?

THE PERFECT COMBINATION

Attempt 3 — take $1\times$ (machine) $+\ 2\times$ (wood):

$$x_1 + 2(x_1 + x_2) \;=\; 3x_1 + 2x_2 \;\le\; 6 + 16 \;=\; 22.$$

This time the combination is *exactly* the objective — no slack wasted. Bound: 22.

And the feasible point $(6, 2)$ *achieves* 22.

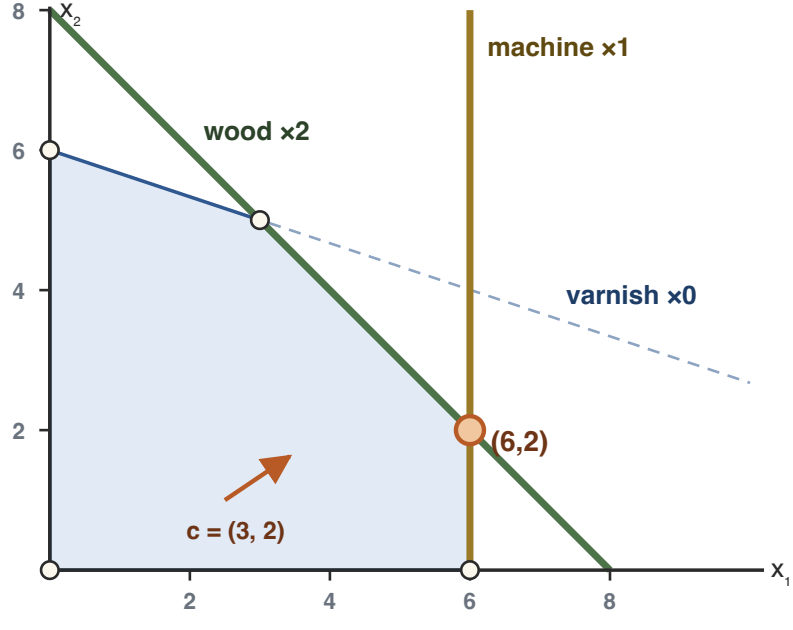
> **CERTIFIED** Every feasible point has value ≤ 22, and $(6, 2)$ has value 22. The pair — point $(6, 2)$, multipliers $(y_1, y_2, y_3) = (1, 2, 0)$ — is a two-line proof of optimality anyone can check.

THE CERTIFICATE, IN THE PICTURE

$y = (1, 2, 0)$: positive weight on **machine** and **wood**, zero on varnish.

Machine and wood are exactly the two lines that *pass through* $(6, 2)$.

Varnish misses $(6, 2)$, and it had to get weight 0. Keep this picture in mind for Part VI.



The constraints in the certificate are the lines through the optimum.

THE BEST BOUND IS AN LP

$$
\begin{array}{lll}
y_1 \times & x_1 \leq 6 & (\text{machine}) \\
y_2 \times & x_1 + x_2 \leq 8 & (\text{wood}) \\
y_3 \times & x_1 + 3x_2 \leq 18 & (\text{varnish}) \\
\text{objective:} & 3x_1 + 2x_2 &
\end{array}
$$

- **Legal:** $y_1, y_2, y_3 \geq 0$ (adding $\leq$-inequalities only works with nonnegative weights).
- **Dominates the objective:** coefficient of x_1: $y_1 + y_2 + y_3 \geq 3$; coefficient of x_2: $y_2 + 3y_3 \geq 2$.
  (Domination suffices because $x_1, x_2 \geq 0$.)
- **Bound obtained:** $6y_1 + 8y_2 + 18y_3$ — make it as small as possible.

THE BEST BOUND IS AN LP

Primal

$$\begin{aligned} \max \quad & 3x_1 + 2x_2 \\ & x_1 \leq 6 \\ & x_1 + x_2 \leq 8 \\ & x_1 + 3x_2 \leq 18 \\ & x_1, x_2 \geq 0 \end{aligned}$$

Dual

$$\begin{aligned} \min \quad & 6y_1 + 8y_2 + 18y_3 \\ & y_1 + y_2 + y_3 \geq 3 \\ & y_2 + 3y_3 \geq 2 \\ & y_1, y_2, y_3 \geq 0 \end{aligned}$$

Three constraints $\rightarrow$ three dual variables; two primal variables $\rightarrow$ two dual constraints.

THE BEST BOUND IS AN LP

Definition The search for the best upper bound is itself a linear program: the **dual** of the original (**primal**) LP.

PART II — THE DUAL RECIPE

RUNNING EXAMPLE, IN MATRICES

$$\max\ 3x_1 + 2x_2 \quad \text{s.t.}$$
$$\underbrace{x_1 \le 6}_{\text{machine}},\quad \underbrace{x_1 + x_2 \le 8}_{\text{wood}},\quad \underbrace{x_1 + 3x_2 \le 18}_{\text{varnish}},\quad x_1, x_2 \ge 0$$

The same LP in matrix form:

$$\max\ \underbrace{\begin{pmatrix}3 & 2\end{pmatrix}}_{c^\intercal}\underbrace{\begin{pmatrix}x_1\\ x_2\end{pmatrix}}_{x} \quad \text{s.t.} \quad \underbrace{\begin{pmatrix}1 & 0\\ 1 & 1\\ 1 & 3\end{pmatrix}}_{A}\begin{pmatrix}x_1\\ x_2\end{pmatrix} \le \underbrace{\begin{pmatrix}6\\ 8\\ 18\end{pmatrix}}_{b},\quad x \ge 0$$

COMBINING CONSTRAINTS

Multiply the constraints by $y_1, y_2, y_3 \geq 0$ and add. The left side is

$$\begin{pmatrix} y_1 & y_2 & y_3 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = (y_1 + y_2 + y_3)\, x_1 + (y_2 + 3y_3)\, x_2.$$

COMBINING CONSTRAINTS

We want the combination to dominate $3x_1 + 2x_2$, coefficient by coefficient:

$$\begin{pmatrix} y_1 & y_2 & y_3 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 3 \end{pmatrix} \geq \begin{pmatrix} 3 & 2 \end{pmatrix}, \quad \text{i.e.} \quad y^{\mathsf{T}} A \geq c^{\mathsf{T}}.$$

Multiplied out: $y_1 + y_2 + y_3 \geq 3$ and $y_2 + 3y_3 \geq 2$ — the dual constraints from Part I.

THE BOUND, IN ONE LINE

Claim If $y \geq 0$ and $y^{\mathsf{T}} A \geq c^{\mathsf{T}}$, then every feasible x satisfies

$$c^{\mathsf{T}} x \;\leq\; y^{\mathsf{T}} A x \;\leq\; y^{\mathsf{T}} b.$$

- First $\leq$: $y^{\mathsf{T}} A$ dominates c^{T} entrywise, and $x \geq 0$.
- Second $\leq$: $Ax \leq b$, weighted by $y \geq 0$.
- Attempt 3 was $y = (1, 2, 0)$: $y^{\mathsf{T}} A = (3, 2) = c^{\mathsf{T}}$, $y^{\mathsf{T}} b = 6 + 16 = 22$.

DUAL OF THE STANDARD FORM

Nothing used the specific numbers. Any LP in Week 9's standard form ($A \in \mathbb{R}^{m \times n}$)

$$\max\ c^{\mathsf{T}} x \quad \text{s.t.} \quad Ax \leq b,\ \ x \geq 0$$

has the same bound: $c^{\mathsf{T}} x \leq y^{\mathsf{T}} b$ for every $y \in \mathbb{R}^m$ with $y \geq 0$ and $y^{\mathsf{T}} A \geq c^{\mathsf{T}}$.

Transposing, $y^{\mathsf{T}} A \geq c^{\mathsf{T}}$ is the same as $A^{\mathsf{T}} y \geq c$: one constraint per column of A.

DUAL OF THE STANDARD FORM

Best bound = smallest right-hand side:

DUAL LP

$$\min\ b^\mathsf{T} y \quad \text{s.t.} \quad A^\mathsf{T} y \geq c,\ \ y \geq 0.$$

Constraints of the primal $\leftrightarrow$ variables of the dual, and vice versa; A becomes A^T; b and c trade places.

MIXED CONSTRAINTS

Real LPs mix $\leq$ rows and $=$ rows:

$$\begin{aligned} \max \quad & x_1 - x_2 \\ \text{s.t.} \quad & x_1 + x_2 = 4 \qquad (R_1) \\ & x_1 \leq 3 \qquad (R_2) \\ & x_1, x_2 \geq 0 \end{aligned}$$

The feasible points form the segment from $(0, 4)$ to $(3, 1)$. On it $x_2 = 4 - x_1$, so the objective is $2x_1 - 4$: the optimum is $\mathbf{2}$, at $(3, 1)$.

Now certify it: combine R_1 and R_2 to prove $x_1 - x_2 \leq 2$.

YOUR TURN: A BOUND OF 2

$$\begin{aligned} y_1 \times \quad & x_1 + x_2 = 4 && (R_1) \\ y_2 \times \quad & x_1 \leq 3 && (R_2) \\ \text{objective:} \quad & x_1 - x_2 \end{aligned}$$

Choose weights and add. The combination must dominate $x_1 - x_2$ on $x \geq 0$, with right-hand side 2.

Hint: with $y_1, y_2 \geq 0$ the best you can prove is 3. What goes wrong? And what may you do to an *equation* that you may not do to an inequality?

THE OBSTRUCTION

With nonnegative weights, $1 \times (R_2)$ gives $x_1 \leq 3$, and since $x_2 \geq 0$, $x_1 - x_2 \leq x_1 \leq 3$. A bound of 3, not 2.

To do better the combination needs coefficient -1 on x_2. The only row containing x_2 is R_1, where it has coefficient $+1$. So R_1 needs weight $y_1 = -1$.

ALLOWED R_1 is an *equation*. Multiplying it by -1 gives $-x_1 - x_2 = -4$, which is just as true. An equation may be scaled by **any** real number; only an inequality cares about the sign.

THE PERFECT COMBINATION, AGAIN

Take $(-1) \times (R_1) + 2 \times (R_2)$:

$$-(x_1 + x_2) + 2x_1 \;=\; x_1 - x_2 \;\leq\; -4 + 6 \;=\; 2.$$

Exactly the objective, bound 2, and $(3, 1)$ achieves 2. Certified.

The equation's weight was negative and nothing broke: scaling an equality never flips anything.

MIXED CONSTRAINTS: THE DUAL

SIGN RULE A $\leq$ row may be multiplied by $y_i \geq 0$ only. An $=$ row may be multiplied by any y_i: its dual variable is **free**.

Everything else is as before. The combination $(y_1 + y_2)\, x_1 + y_1\, x_2 \leq 4y_1 + 3y_2$ must dominate $x_1 - x_2$ coefficient by coefficient, and we want the smallest right-hand side:

$$\min\ 4y_1 + 3y_2 \quad \text{s.t.}$$
$$y_1 + y_2 \geq 1,\ \ y_1 \geq -1,\ \ y_1 \text{ free},\ y_2 \geq 0$$

$y^\star = (-1, 2)$ is feasible, with value $-4 + 6 = 2$: the bounds meet.

THE RECIPE

For $\max\ c^{\mathsf{T}}x$ with rows $a_i^{\mathsf{T}}x \le b_i$ or $a_i^{\mathsf{T}}x = b_i$, and $x \ge 0$:

1. Give row i a weight y_i: $y_i \ge 0$ for a $\le$ row, y_i free for an $=$ row.
2. Add the rows: $y^{\mathsf{T}}Ax \le y^{\mathsf{T}}b$ for every feasible x.
3. Dominate the objective, one variable at a time: $(A^{\mathsf{T}}y)_j \ge c_j$ for every j (this uses $x \ge 0$). Then $c^{\mathsf{T}}x \le y^{\mathsf{T}}Ax \le y^{\mathsf{T}}b$.
4. Minimise the bound $b^{\mathsf{T}}y$. That is the dual.

This is all max-flow will need. (A $\ge$ row works too: multiply it by $y_i \le 0$, which turns it into a $\le$ row.)

DUAL OF THE DUAL

FACT For standard form, the dual of the dual is the primal: negate the dual to get a standard-form max, take its dual, negate back.

$$\begin{aligned}
&\min\{b^\mathsf{T} y : A^\mathsf{T} y \geq c,\ y \geq 0\} \\
&= -\max\{-b^\mathsf{T} y : -A^\mathsf{T} y \leq -c,\ y \geq 0\} \\
&\xrightarrow{\text{dual}} \quad -\min\{-c^\mathsf{T} x : -Ax \geq -b,\ x \geq 0\} \\
&= \max\{c^\mathsf{T} x : Ax \leq b,\ x \geq 0\}.
\end{aligned}$$

DUAL OF THE DUAL

Consequences:

- Duality is a genuine *pairing*, not a one-way construction — each LP certifies the other.
- "Primal" and "dual" are roles, not properties: solve or analyse whichever of the pair is more convenient.

PART III — WEAK AND STRONG DUALITY

WEAK DUALITY

THEOREM (WEAK DUALITY) If x is feasible for the primal $\max\{c^\mathsf{T} x : Ax \le b,\ x \ge 0\}$ and y is feasible for its dual, then $c^\mathsf{T} x \le b^\mathsf{T} y$.

Proof (one line):

$$c^\mathsf{T} x \ \le\ (A^\mathsf{T} y)^\mathsf{T} x \ =\ y^\mathsf{T}(Ax) \ \le\ y^\mathsf{T} b. \qquad \blacksquare$$

Note: the proof is exactly the "combine the constraints" trick from Part I.

WEAK DUALITY

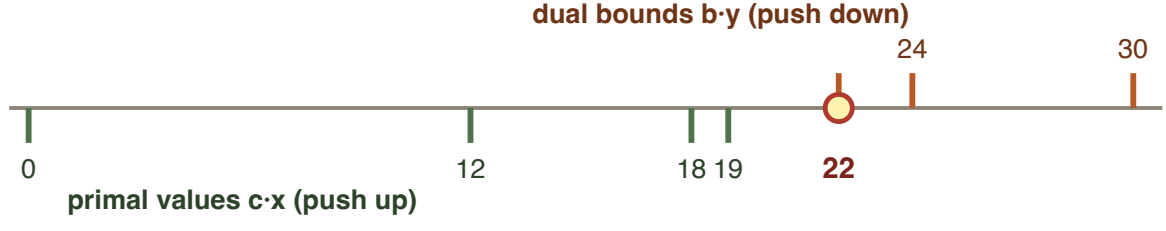
Unpacking each step of $c^\mathsf{T} x \leq (A^\mathsf{T} y)^\mathsf{T} x = y^\mathsf{T}(Ax) \leq y^\mathsf{T} b$:

- **First $\leq$:** $A^\mathsf{T} y \geq c$ coefficient-wise (dual feasibility), and $x \geq 0$ — multiplying a bigger coefficient by a nonnegative number can only increase the sum.
- **Middle $=$:** just re-bracketing, $(A^\mathsf{T} y)^\mathsf{T} x = y^\mathsf{T} A x = y^\mathsf{T}(Ax)$.
- **Second $\leq$:** $Ax \leq b$ row-wise (primal feasibility), and $y \geq 0$.

CERTIFICATES FOR FREE

COROLLARY If $x^\star$ is primal feasible, $y^\star$ is dual feasible, and $c^\mathsf{T} x^\star = b^\mathsf{T} y^\star$, then **both are optimal**.

Every primal value sits below every dual value — if they touch, neither side can move.



The running example: vertex values $0, 12, 18, 19$ from below; Attempts 1–3 give bounds $30, 24, 22$ from above; they meet at 22.

This is the same logic as Week 8's lecture: a flow of value v plus a cut of capacity v certify each other.

STRONG DUALITY

Theorem (strong duality — stated, not proved) If the primal has a finite optimum, so does the dual, and the two optimal values are **equal**:

$$\begin{aligned} &\max\{c^{\mathsf{T}}x : Ax \le b,\ x \ge 0\} \\ =\ &\min\{b^{\mathsf{T}}y : A^{\mathsf{T}}y \ge c,\ y \ge 0\}. \end{aligned}$$

So the meeting at 22 was no accident: *a perfect certificate always exists.* The bound-combining game can always be played to a draw.

THE FEASIBILITY CASES

Each LP has an optimum, is unbounded, or is infeasible. Of the nine pairings, only four occur:

PRIMAL \ DUAL	OPTIMUM	UNBOUNDED	INFEASIBLE
optimum	**possible** values equal	**never** weak duality	**never** strong duality
unbounded	**never** weak duality	**never** weak duality	**possible** forced by weak duality
infeasible	**never** strong duality	**possible** forced by weak duality	**possible** example next

A dual-feasible y caps every primal value at $b^{\mathsf{T}} y$ — so an unbounded side forces the other to be infeasible.

THE FEASIBILITY CASES

Both infeasible at once:

Primal

$$\begin{aligned} \max \quad & x_1 \\ & x_1 - x_2 \leq -1 \\ & -x_1 + x_2 \leq -1 \\ & x_1, x_2 \geq 0 \end{aligned}$$

Dual

$$\begin{aligned} \min \quad & -y_1 - y_2 \\ & y_1 - y_2 \geq 1 \\ & -y_1 + y_2 \geq 0 \\ & y_1, y_2 \geq 0 \end{aligned}$$

- Primal: add the two rows to get $0 \leq -2$. Infeasible.
- Dual: add the two rows to get $0 \geq 1$. Infeasible.

PART IV — MAX-FLOW / MIN-CUT, ROUND TWO

RECALL WEEK 8

> **THEOREM (MAX-FLOW = MIN-CUT)** In any s–t network, the maximum flow value equals the minimum capacity of an s–t cut.

Week 8's proof: run Edmonds–Karp (Ford–Fulkerson with BFS) until no augmenting path exists; let S = vertices reachable from s in the *residual graph*; the cut $(S, V \setminus S)$ is saturated, so its capacity equals the flow value.

Today we re-prove the same theorem *without ever mentioning residual graphs* — it will fall out of LP duality.

MAX FLOW AS AN LP

Digraph $G = (V, E)$, capacities $c_e \geq 0$, source s, sink t (no edges into s). One variable f_e per edge; $\delta^+(v)$ = edges out of v, $\delta^-(v)$ = edges into v:

$$\begin{aligned} \max \quad & \textstyle\sum_{e \in \delta^+(s)} f_e \\ \text{s.t.} \quad & f_e \leq c_e \quad \forall e \in E \\ & \textstyle\sum_{e \in \delta^-(v)} f_e \; - \; \sum_{e \in \delta^+(v)} f_e = 0 \quad \forall v \neq s, t \\ & f_e \geq 0 \quad \forall e \in E \end{aligned}$$

A maximisation with $\leq$ rows and $=$ rows. Give each row a weight:

- capacity constraint of edge $e \rightarrow$ dual variable $y_e \geq 0$
- conservation constraint at vertex $v \rightarrow$ dual variable z_v, *free* (it's an equality)

DERIVING THE DUAL

Multiply every row by its weight and add, as in Part II:

$$\sum_e y_e f_e \;+\; \sum_{v \neq s,t} z_v \Big(\sum_{e \in \delta^-(v)} f_e - \sum_{e \in \delta^+(v)} f_e \Big) \;\leq\; \sum_e c_e y_e .$$

Then collect the coefficient of each f_e on the left. Since $f_e \geq 0$, it must dominate the objective's coefficient of f_e: 1 if e leaves s, else 0. **That is the dual constraint for edge e.**

ONE EDGE AT A TIME

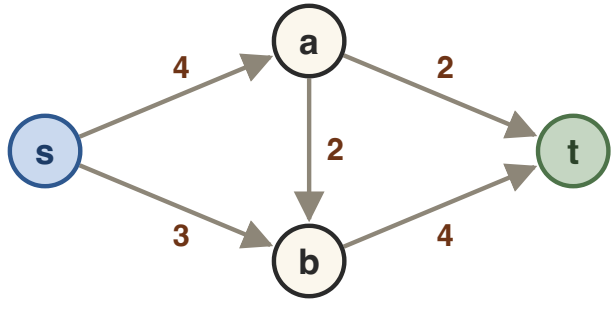
Take an edge $e = (u, v)$. Every row that contains f_e:

ROW CONTAINING f_e	ADDS TO ITS COEFFICIENT
capacity row of e	$+y_e$
conservation at u **(**e **leaves** u**)**	$-z_u$
conservation at v **(**e **enters** v**)**	$+z_v$

Dominate the objective coefficient, $[u = s]$: $y_e - z_u + z_v \geq [u = s]$, i.e. $y_e \geq [u = s] + z_u - z_v$, with no z_s or z_t term (s and t have no conservation row).

ONE EDGE AT A TIME

Set $z_s := 1$ and $z_t := 0$. Then every edge, including those at s and t, reads the same way: $y_e \geq z_u - z_v$.



Week 9's network.

$$
\begin{aligned}
sa: &\quad y_{sa} \geq 1 - z_a \\
sb: &\quad y_{sb} \geq 1 - z_b \\
ab: &\quad y_{ab} \geq z_a - z_b \\
at: &\quad y_{at} \geq z_a \\
bt: &\quad y_{bt} \geq z_b
\end{aligned}
$$

Five edges, five dual constraints. The right-hand side of the combination is $\sum_e c_e\, y_e$: minimise it.

DERIVING THE DUAL

DUAL LP

$$\begin{aligned} \min \quad & \textstyle\sum_e c_e\, y_e \\ \text{s.t.} \quad & y_e \;\geq\; z_u - z_v \quad \forall e = (u,v) \in E \\ & z_s = 1, \quad z_t = 0, \quad y_e \geq 0 \end{aligned}$$

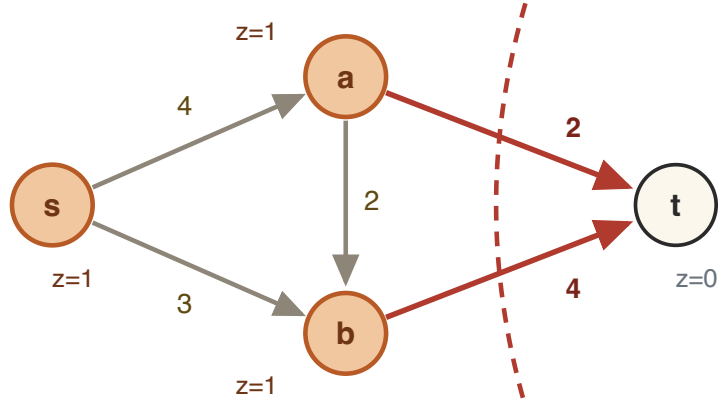
Intuition: pay c_e per unit of y_e; y_e must cover any *drop in the label* z along edge e; s is labelled 1, t is labelled 0.

READING THE DUAL: IT PRICES CUTS

Try a $0/1$ solution: pick $S \subseteq V$ with $s \in S, t \notin S$, and set $z_v = 1$ if $v \in S$, else 0.

Then $y_e \geq z_u - z_v$ forces $y_e \geq 1$ exactly on edges *leaving* S — and $y_e = 0$ works everywhere else.

Dual objective $= \sum_{e \text{ leaving } S} c_e$ = **the capacity of the cut S.**



Week 9's network. $S = \{s, a, b\}$: $z = 1$ on S, $z_t = 0$; $y_e = 1$ on the two red edges. Value $2 + 4 = 6$ — exactly last week's max flow. ($S = \{s\}$ gives $4 + 3 = 7$: a valid bound, but not tight.)

ROUNDING TO A CUT

Every cut is a $0/1$ dual solution, so the LP optimum is at most the min cut. For the other direction, take any optimal dual (y, z) and **round it to a cut** that costs no more.

- **Clip:** raise each negative z_v to 0, lower each $z_v > 1$ to 1.
  Still feasible: each new gap $z_u - z_v \leq \max(0, \text{old gap}) \leq y_e$.
- **Threshold:** pick θ uniformly from $(0, 1]$ and let

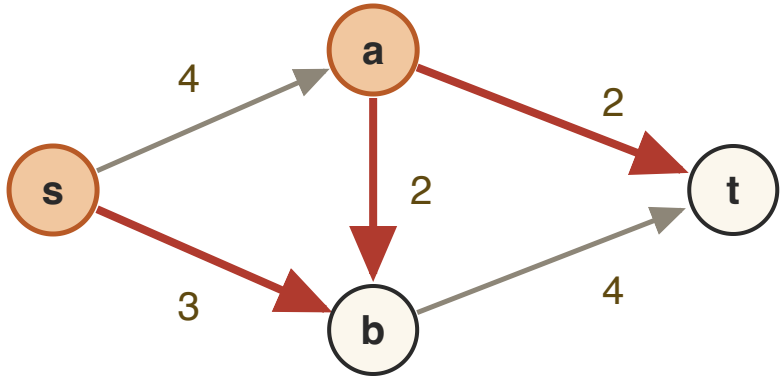
$$S_\theta = \{v : z_v \geq \theta\}.$$

  It contains s ($z_s = 1$) and never t ($z_t = 0$): an s–t cut.

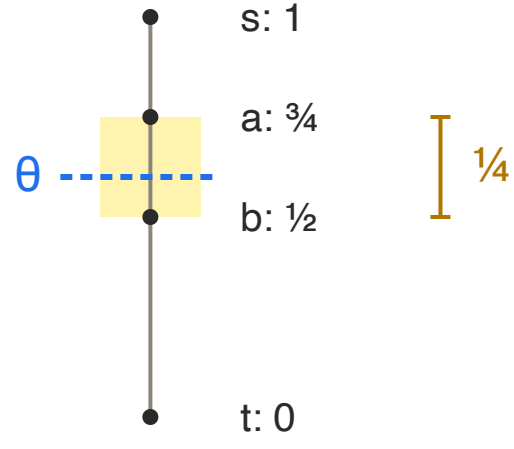
THRESHOLD ROUNDING: A PICTURE

A feasible fractional dual: $z = (1, \frac{3}{4}, \frac{1}{2}, 0)$ on (s, a, b, t), with $y_{uv} = \max(0, z_u - z_v)$.

Capacities on edges



Vertex potentials



Orange: S_θ. Red: edges leaving S_θ.

Shaded interval: a → b crosses

Move θ: 0.60

S_θ = {s, a}, cut capacity = 7.

Edge $a \to b$ crosses exactly when $\frac{1}{2} < \theta \leq \frac{3}{4}$, so it crosses with probability $\frac{1}{4} = z_a - z_b$.

ROUNDING TO A CUT

For the proof, return to an optimal dual (y, z) after clipping.

- Edge $e = (u, v)$ leaves S_θ exactly when $z_v < \theta \le z_u$:

$$\Pr_\theta[e \text{ leaves } S_\theta] = \max(0,\ z_u - z_v) \ \le\ y_e.$$

- Sum over edges, weighted by capacity:

$$\begin{aligned}\mathbb{E}_\theta\big[\mathrm{cap}(S_\theta)\big] &= \textstyle\sum_e c_e \Pr_\theta[e \text{ leaves } S_\theta] \\ &\le \textstyle\sum_e c_e\, y_e \ =\ \text{LP optimum}.\end{aligned}$$

- Some θ does at least as well as the average: a cut of capacity $\le$ LP optimum.

ROUNDING TO A CUT

Cuts bound the LP from above, and rounding finds a cut no bigger than it. So the dual LP optimum *is* the min cut.

MAX-FLOW = MIN-CUT

$$\text{max flow} \overset{\text{strong duality}}{=} \text{dual LP optimum} \overset{\text{rounding}}{=} \text{min cut}.$$

SAME THEOREM, TWO PROOFS

	Week 8: residual graph	Today: LP duality
what you build	an algorithm (Ford–Fulkerson)	a certificate framework, no algorithm
integrality	integer capacities $\Rightarrow$ integer flow (integer augmentations)	$0/1$ cut labels via dual rounding, even for real capacities
scope	flows and close relatives	**every** linear program
the certificate	the cut found by the last BFS	any optimal dual solution

SAME THEOREM, TWO PROOFS

TAKEAWAY The residual-graph proof generalises to **algorithms** (Ford–Fulkerson); the LP proof generalises to **every LP**. Knowing both, you can pick the lens that fits the problem in front of you.

PART V — THE MATCHING CAPSTONE: KŐNIG'S THEOREM

THE MATCHING THREAD SO FAR

Week	What we did with matchings
W3	maximum matching on *trees* by DP
W4	greedy matching on trees + exchange arguments
W8	maximum *bipartite* matching via max flow
W9	the matching LP relaxation; integrality gaps
W10	**today: the capstone — duality explains it all**

The question left hanging since Week 8: when we found a maximum bipartite matching, *what certified that we could stop?*

DERIVING THE COVER DUAL

Graph $G = (V, E)$; variable x_e per edge (how much of e we take). The matching LP:

$$\max \sum_{e \in E} x_e \quad \text{s.t.} \quad \sum_{e \ni v} x_e \leq 1 \;\; \forall v \in V, \quad x \geq 0.$$

Give each vertex row a weight y_v. It is a $\leq$ row, so $y_v \geq 0$. Multiply and add:

$$\sum_v y_v \sum_{e \ni v} x_e \;\leq\; \sum_v y_v.$$

Then collect the coefficient of each x_e on the left: it must be ≥ 1, the objective's coefficient.

ONE EDGE AT A TIME

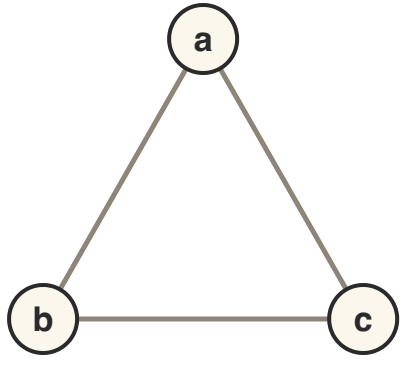
Take an edge $e = uv$. Every row that contains x_e:

ROW CONTAINING x_e	ADDS TO ITS COEFFICIENT
vertex row of u	$+y_u$
vertex row of v	$+y_v$
any other vertex row	nothing: e doesn't touch it

Coefficient $y_u + y_v \geq 1$.
The right-hand side is $\sum_v y_v$: minimise it.

ONE EDGE AT A TIME

Week 9's triangle: three vertex rows give $y_a, y_b, y_c \geq 0$; three edges give three dual constraints.



$$\begin{aligned} \min \quad & y_a + y_b + y_c \\ ab: \quad & y_a + y_b \geq 1 \\ ac: \quad & y_a + y_c \geq 1 \\ bc: \quad & y_b + y_c \geq 1 \end{aligned}$$

Each edge must be covered by its two endpoints: a **fractional vertex cover**.

MATCHING $\leftrightarrow$ VERTEX COVER

Matching LP (primal)

$$\begin{aligned} \max \quad & \textstyle\sum_{e \in E} x_e \\ & \textstyle\sum_{e \ni v} x_e \le 1 \quad \forall v \in V \\ & x_e \ge 0 \end{aligned}$$

Its dual (derived)

$$\begin{aligned} \min \quad & \textstyle\sum_{v \in V} y_v \\ & y_u + y_v \ge 1 \quad \forall uv \in E \\ & y_v \ge 0 \end{aligned}$$

The dual is the **fractional vertex cover** LP: pay for vertices so that every edge is covered at least once. The two problems we kept meeting are *LP duals*.

WEAK DUALITY, COMBINATORIALLY

OBSERVATION For *any* matching M and *any* vertex cover C in *any* graph: $|M| \leq |C|$.

Why: each edge of M must contain at least one vertex of C; and no vertex of C can account for two edges of M, since matching edges are disjoint. So C has at least one private vertex per matching edge. ■

This is exactly the weak duality inequality $c^{\mathsf{T}} x \leq b^{\mathsf{T}} y$, specialised to $0/1$ solutions of the two LPs.

WEAK DUALITY, COMBINATORIALLY

So: a matching and a cover *of the same size* would certify each other. Do they always exist? In general graphs, no (triangle: max matching 1, min cover 2). But…

BIPARTITE: THE GAP CLOSES

Fact (integrality — Week 9) If G is **bipartite**, the matching LP and the vertex cover LP both have optimal solutions that are $0/1$-valued.

(Week 9: the incidence matrix of a bipartite graph is *totally unimodular*, so by Cramer's rule every polytope vertex is integral.)

BIPARTITE: THE GAP CLOSES

Recall Week 9's warning example: on the triangle C_3, setting $x_e = \frac{1}{2}$ on all three edges gives matching-LP value $\frac{3}{2} > 1$ — an integrality gap. Odd cycles are the culprit, and bipartite graphs have none.

Now chain the equalities:

$$\begin{aligned} \max \text{ matching} &\overset{\text{integrality}}{=} \mathrm{LP}_{\text{match}} \\ &\overset{\text{strong duality}}{=} \mathrm{LP}_{\text{cover}} \\ &\overset{\text{integrality}}{=} \min \text{ vertex cover}. \end{aligned}$$

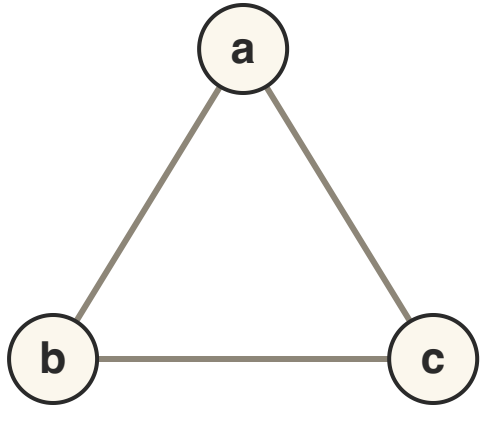
KŐNIG'S THEOREM

Theorem (Kőnig, 1931) In every bipartite graph, the size of a maximum matching equals the size of a minimum vertex cover.

- The chain on the previous slide *is* the proof — duality plus integrality, no new combinatorics needed.
- Practical upshot: a matching M and a cover C with $|M| = |C|$ form a **mutual certificate** — hand both to the skeptic.

THE TRIANGLE

Which equalities from the bipartite chain survive on K_3?



Compute all four optima:

1. Maximum matching
2. Maximum fractional matching
3. Minimum fractional cover
4. Minimum vertex cover

THE TRIANGLE: DUALITY SURVIVES

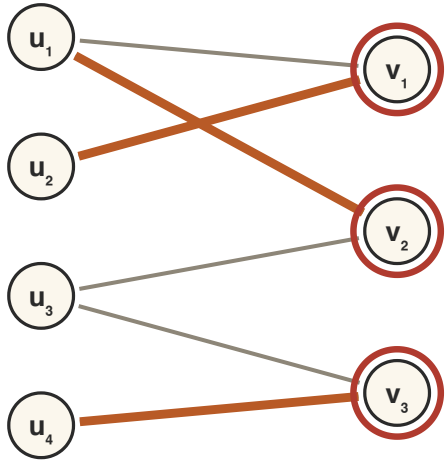
$$\underbrace{1}_{\substack{\max \\ \text{matching}}} < \underbrace{\tfrac{3}{2}}_{\text{matching LP}} = \underbrace{\tfrac{3}{2}}_{\text{cover LP}} < \underbrace{2}_{\substack{\min \\ \text{cover}}}.$$

Put $x_e = \frac{1}{2}$ on every edge and $y_v = \frac{1}{2}$ on every vertex. Both LP solutions are feasible and have value $\frac{3}{2}$, so weak duality certifies that both are optimal.

Any two edges share a vertex, so a matching has size at most 1. A vertex cover needs 2 vertices.

> **WHAT BREAKS?** Strong duality still gives the middle equality. Both **integrality** equalities fail.

WORKED EXAMPLE: A MATCHING CERTIFIES A COVER



Orange edges: matching M. Red rings: cover $C = \{v_1, v_2, v_3\}$.

Bipartite graph: edges $u_1v_1, u_1v_2, u_2v_1, u_3v_2, u_3v_3, u_4v_3$.

Matching
$M = \{u_1v_2,\ u_2v_1,\ u_4v_3\}$, size $\mathbf{3}$ (only u_3 unmatched).

Cover $C = \{v_1, v_2, v_3\}$, size $\mathbf{3}$: every edge ends on the right, so all six are covered.

WORKED EXAMPLE: A MATCHING CERTIFIES A COVER

MUTUAL CERTIFICATION Any cover needs $\geq |M| = 3$ vertices, so C is a minimum cover; any matching has $\leq |C| = 3$ edges, so M is a maximum matching. Each is the other's proof.

THE VERTEX COVER ARC

Graph	Min vertex cover	When
trees	easy — DP / Gallai	Week 3
bipartite	polynomial — Kőnig: reduce to matching	Week 10 (today)
general graphs	…what do you think?	Week 11

Duality gave us the bipartite case as a two-line corollary. For general graphs the matching–cover gap reopens (remember the triangle) — and something much deeper goes wrong. That story is **NP-completeness, next week**.

PART VI — COMPLEMENTARY SLACKNESS

COMPLEMENTARY SLACKNESS

When do a primal-feasible x and dual-feasible y actually meet, $c^\mathsf{T} x = b^\mathsf{T} y$?

THEOREM (COMPLEMENTARY SLACKNESS) Feasible x and y are *both optimal* if and only if:

- for every j: $x_j > 0 \;\Rightarrow\; (A^\mathsf{T} y)_j = c_j$ (used variable $\Rightarrow$ its dual constraint is tight)
- for every i: $y_i > 0 \;\Rightarrow\; (Ax)_i = b_i$ (paid-for constraint $\Rightarrow$ it is tight)

COMPLEMENTARY SLACKNESS

Slogan: at the optimum, nothing is both *used* and *slack*. You never pay a positive price y_i for a constraint you aren't pressing against, and you never run a positive activity x_j whose dual price exceeds its value.

WHY IT HOLDS: SQUEEZE THE WEAK DUALITY CHAIN

Recall the one-line proof:

$$c^{\mathsf{T}} x \overset{(1)}{\leq} y^{\mathsf{T}} A x \overset{(2)}{\leq} y^{\mathsf{T}} b.$$

If x and y are both optimal, strong duality says the two ends are *equal* — so (1) and (2) are equalities.

WHY IT HOLDS: SQUEEZE THE WEAK DUALITY CHAIN

Both $\leq$ in $c^\mathsf{T}x \overset{(1)}{\leq} y^\mathsf{T}Ax \overset{(2)}{\leq} y^\mathsf{T}b$ are equalities at an optimal pair. Now read each termwise.

Inequality (1) written termwise: $\sum_j x_j\big((A^\mathsf{T}y)_j - c_j\big) = 0$, a sum of *nonnegative* terms. A nonnegative sum is zero only if every term is zero: $x_j > 0$ forces $(A^\mathsf{T}y)_j = c_j$.

Same for (2): $\sum_i y_i\big(b_i - (Ax)_i\big) = 0$ gives $y_i > 0 \Rightarrow (Ax)_i = b_i$. And conversely, if all terms vanish the chain collapses to equality, so both are optimal. ■

WORKED CHECK: THE RUNNING EXAMPLE

Claimed optimal pair: $x^\star = (6, 2)$ and $y^\star = (1, 2, 0)$. First, the two primal variables:

CONDITION	CHECK	
$x_1 = 6 > 0$	dual constraint 1 tight? $y_1 + y_2 + y_3 = 1 + 2 + 0 = 3 = c_1$	✓
$x_2 = 2 > 0$	dual constraint 2 tight? $y_2 + 3y_3 = 2 + 0 = 2 = c_2$	✓

WORKED CHECK: THE RUNNING EXAMPLE

Then the three dual variables, one per resource:

CONDITION	CHECK	
$y_1 = 1 > 0$	machine tight? $x_1 = 6 = b_1$	✓
$y_2 = 2 > 0$	wood tight? $x_1 + x_2 = 6 + 2 = 8 = b_2$	✓
$y_3 = 0$	varnish is slack, $x_1 + 3x_2 = 12 < 18$ — allowed only because $y_3 = 0$	✓

WORKED CHECK: THE RUNNING EXAMPLE

All conditions hold, and both points are feasible $\Rightarrow$ both optimal — *without computing either objective value*.

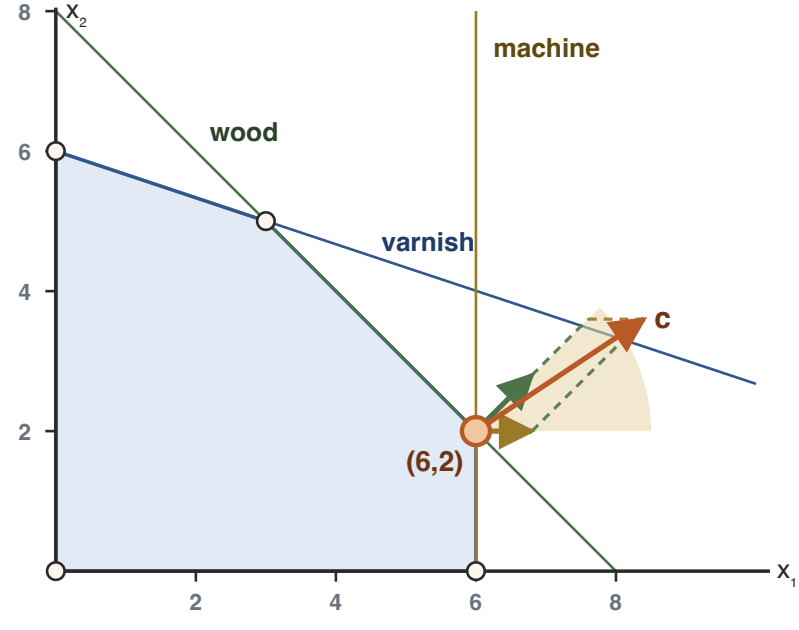
The varnish row is the one with content: a constraint with slack *must* get multiplier 0. Attempt 1 put weight on varnish, and that is exactly why it could not reach 22.

THE CHECK, IN THE PICTURE

At $(6, 2)$ the tight rows are machine and wood. Draw their normals, $(1, 0)$ and $(1, 1)$, at the point.

Both $x_j > 0$, so both dual constraints are tight: $y^\mathsf{T} A = c^\mathsf{T}$. That says c is a combination of the row normals: $c = 1 \cdot (1, 0) + 2 \cdot (1, 1)$, so $y^\star = (1, 2, 0)$.

c lies *inside* the cone of the two normals, so both weights are positive. Varnish is slack; it gets 0 and plays no part.



The prices are the coordinates of c in the cone of the tight constraints' normals.

SOLVING FOR THE CERTIFICATE

Given only the claimed optimum $x^\star = (6, 2)$, complementary slackness hands us equations for $y^\star$:

- varnish is slack $\Rightarrow y_3 = 0$;
- $x_2 > 0 \Rightarrow y_2 + 3y_3 = 2$, so $y_2 = 2$;
- $x_1 > 0 \Rightarrow y_1 + y_2 + y_3 = 3$, so $y_1 = 1$.

Check $y^\star = (1, 2, 0) \geq 0$: it is dual feasible, so $(6, 2)$ is certified. (These equations pin $y^\star$ down uniquely, so a negative entry would have meant $(6, 2)$ is not optimal.)

WHEN A PRICE GOES NEGATIVE

Try complementary slackness at a vertex that is *not* optimal: $x = (3, 5)$, value 19. Wood and varnish are tight; the machine is slack.

- machine slack $\Rightarrow y_1 = 0$;
- $x_1, x_2 > 0 \Rightarrow y_2 + y_3 = 3$ and $y_2 + 3y_3 = 2$;
- so $y_2 = \frac{7}{2}$ and $y_3 = -\frac{1}{2}$.

The combination $\frac{7}{2}\,(\text{wood}) - \frac{1}{2}\,(\text{varnish})$ gives exactly 19 — but subtracting a $\leq$ constraint is not allowed. **No certificate exists at $(3, 5)$.**

WHEN A PRICE GOES NEGATIVE

The negative price on varnish says: *loosening varnish pays*.

- Keep wood tight and move off varnish: trade one table for one chair, $(3, 5) \to (4, 4)$. Varnish loosens by 2, and the objective gains $(-\frac{1}{2})(-2) = +1$.
- Continue until the machine becomes tight: $(6, 2)$, value 22.
- There the prices are $(1, 2, 0) \geq 0$: a valid certificate, so stop.

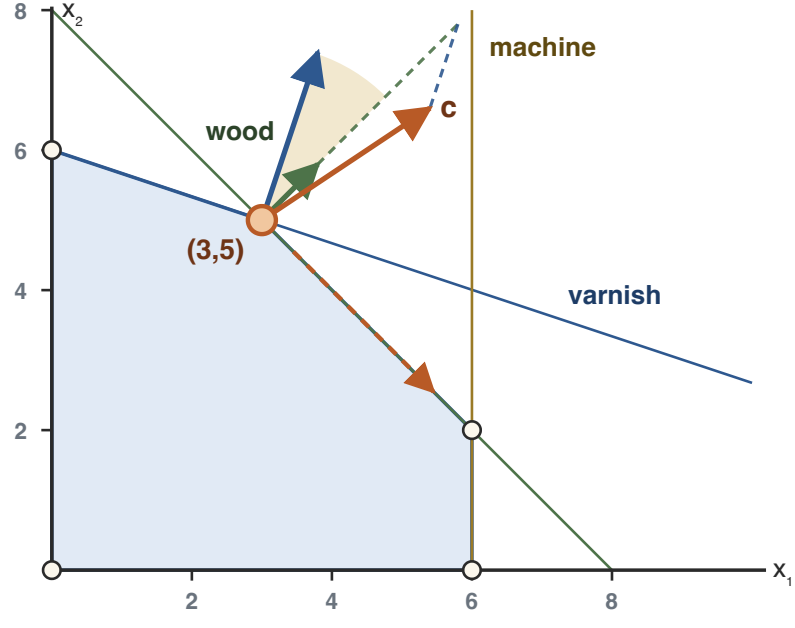
This illustrates how the **simplex method** uses prices to choose pivots. To certify a feasible point, check **dual feasibility** ($y \geq 0$ and $A^{\mathsf{T}} y \geq c$) together with complementary slackness. (Week 12's *primal–dual* algorithms use complementary slackness as a design principle too.)

A NEGATIVE PRICE, IN THE PICTURE

At $(3, 5)$ the tight rows are wood and varnish. Their normals, $(1, 1)$ and $(1, 3)$, are drawn at the point.

$c = \frac{7}{2}(1, 1) - \frac{1}{2}(1, 3)$. The only way to write c uses a negative weight: c lies *outside* the cone, on the far side of the wood normal.

So sliding along the wood edge, away from varnish, moves in the direction of c. That is the improving move to $(6, 2)$.



A vertex is optimal exactly when c lies in the cone of its tight constraints' normals.

TAKEAWAYS

- Every LP has a **dual**: the LP of best bounds obtained by combining constraints. Weights on $\leq$ rows are ≥ 0, weights on $=$ rows are free, and the combination must dominate the objective coefficient by coefficient.
- **Weak duality** ($c^{\mathsf{T}} x \leq b^{\mathsf{T}} y$) is a one-line proof; **strong duality** says every finite primal optimum has a matching dual certificate.
- **Max-flow = min-cut** is LP duality: the dual of the flow LP prices cuts. Two proofs, two generalisations — algorithms vs. all of LP.

TAKEAWAYS

- **Kőnig:** in bipartite graphs, max matching $=$ min vertex cover — duality $+$ integrality; matching and cover certify each other.
- **Complementary slackness:** nothing optimal is both used and slack. In the picture, c sits inside the cone of the tight constraints' normals, and the prices are its coordinates there. A negative price points to an improving move.